# Dev Patel

Software Engineer — Backend, Full-Stack & Applied AI
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## SUMMARY

Software engineer with 1.5 years of enterprise experience (12 months full-time plus a 5-month internship) building ASP.NET Core and Angular applications, REST APIs, and SQL workflows for two Indian government platforms. Cut data-retrieval time about 30% through query and index tuning. Built and deployed a full-stack RAG document assistant with Next.js, FastAPI, Hugging Face, and Pinecone. M.S. Computer Science at UT Arlington, graduating May 2027; available for full-time roles from June 2027.

## TECHNICAL SKILLS

**Languages:** C#, Python, TypeScript, JavaScript, SQL, PowerShell
**Backend:** ASP.NET Core, REST APIs, FastAPI, LINQ, MVC, API design, error handling
**Frontend:** Angular, Next.js, HTML5, CSS3
**Databases:** SQL Server, MySQL, stored procedures, indexing, query optimization, data modeling, RDLC reports
**AI / ML:** RAG, Hugging Face, Pinecone (vector search), OpenCV, feature matching (SIFT, ORB, SuperPoint, LightGlue)
**Cloud & DevOps:** Git, GitHub, Azure DevOps, CI/CD pipelines, Vercel, Render
**Practices:** Agile/Scrum, unit testing, API testing (Postman), code reviews, debugging

## EXPERIENCE

**Software Developer (.NET)** — Dev Information Technology Pvt. Ltd., Jaipur, India — Jun 2024 – Jun 2025

- Built backend services and REST APIs in ASP.NET Core and C# for two government platforms, Kaushal Darpan (Rajasthan education) and NICSI (Government of India IT services).
- Cut data-retrieval time about 30% by rewriting slow MySQL queries and stored procedures and adding targeted indexes.
- Delivered student-records, fee-management, and departmental-workflow modules end to end, from API to Angular UI.
- Built RDLC reports that gave administrators operational data for decisions.
- Shipped in Agile/Scrum sprints with frontend, QA, and product teams; reviewed teammates' code and fixed production defects.

**Software Developer Intern** — Dev Information Technology Pvt. Ltd., Jaipur, India — Jan 2024 – May 2024

- Implemented backend business logic and Angular screens for enterprise modules under senior developer guidance.
- Tested and debugged APIs with Postman and shipped releases through Azure DevOps CI/CD pipelines.

## PROJECTS

**AI-Powered RAG Document Assistant** — Next.js, FastAPI, Python, Hugging Face, Pinecone — Live demo · Code

- Built a full-stack app where users upload PDF or TXT files and ask questions answered from the most relevant passages.
- Embedded document chunks with Hugging Face models and stored them in Pinecone for vector search; served retrieval and answers through a FastAPI backend.
- Deployed the Next.js frontend on Vercel and the API on Render with CI/CD from GitHub.

**Multi-View Geometry: Classical vs. Deep Feature Matching** — Python, OpenCV, SuperPoint, LightGlue

- Benchmarked SIFT and ORB against SuperPoint and LightGlue on 5,600+ frames of KITTI driving data.
- Estimated fundamental matrices with RANSAC filtering and recovered camera pose from the matched features.

## EDUCATION

**M.S., Computer Science** — University of Texas at Arlington — 2025 – May 2027 (expected)

Coursework: Advanced Software Engineering, Data Structures & Algorithms, Database Systems, Cloud Computing & Big Data, Data Mining, Artificial Intelligence, Computer Vision

**B.Tech, Information Technology** — Ganpat University, Gujarat, India — 2020 – 2024

## CERTIFICATIONS

Complete C# Masterclass (Udemy) · Python and Machine Learning (DevTown)